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A pullback-corrected scalar auxiliary variable optimizer with momentum and adaptive mobility

Jiahao Zhang[a,1], Shiheng Zhang[b,1], Guang Lin[a,c,*]

[a] *School of Mechanical Engineering, Purdue University, West Lafayette, IN 47906, USA*
[b] *Department of Mathematics, University of Washington, Seattle, WA 98195, USA*
[c] *Department of Mathematics, Purdue University, West Lafayette, IN 47906, USA*

Abstract

Objectives in scientific machine learning are often prescribed as a sum of several terms, such as the residual, boundary, initial, and data losses of a physics-informed neural network. In the pullback-corrected scalar auxiliary variable (PB–SAV) method, one scalar tracks the shifted objective while the component gradients build a positive semidefinite curvature correction of rank at most the number of components. We carry that correction into an optimizer with momentum and an adaptive mobility, applying it to the gradient and the stored momentum in a single implicit solve. A mobility that is nonincreasing in the Loewner order yields an exact modified energy law, covering Euclidean and AMSGrad-type choices; the corresponding identity for momentum appended after the solve carries a cross term of indefinite sign. For a fixed mobility we give a necessary and sufficient condition for local stability at a stationary point, depending on the Hessian minus twice the correction, and show that it also gives local geometric convergence for every scalar relaxation sequence. The implicit solve reduces to a dense system whose order is the number of components. In the forward Burgers comparison, four components reduce the mean tail objective by 64.7% and the final solution error by 50.2% relative to one component at the same learning rate and momentum settings.

Keywords: scalar auxiliary variable, energy stability, momentum, adaptive optimization, Gauss–Newton, physics-informed neural network, operator learning

*Corresponding author (guanglin@purdue.edu)
[1] These authors contributed equally to this work.

1. Introduction

Many scientific machine learning problems have objectives with m prescribed components,

$$F(\theta) = \sum_{i=1}^{m} E_i(\theta). \tag{1}$$

For example, physics-informed neural network (PINN) objectives may combine residual, boundary, initial-condition, and data terms [1, 2], with regularization added when needed. In operator learning such as DeepONet [3] , a data term can likewise be separated from weight decay. These components can have different scales and gradient directions. Once their gradients are summed, part of this directional information may be hidden by aggregation or cancellation. Imbalance between the residual and boundary terms of a PINN objective, and its effect on training, motivate the adaptive loss weighting developed for physics-informed models [4, 5, 6]. Those methods rescale the components before they are summed. The method developed here keeps the sum in Eq. (1) as it stands and uses the component gradients to build the curvature inside the solve.

Most first-order optimization methods act primarily on the aggregated gradient. Momentum incorporates information from previous iterations [7], while adaptive methods such as Adam and AMSGrad introduce coordinate-wise scaling [8, 9]. Neither mechanism directly uses the prescribed component structure of the objective. In an energy-dissipative scheme, momentum adds cross-iteration terms and adaptive scaling changes the quadratic form in which those terms are measured.

The scalar auxiliary variable (SAV) approach constructs energy-stable schemes [10, 11]. Originally developed for dissipative gradient flows, SAV and its relaxed variants introduce auxiliary variables so that a modified energy satisfies a discrete dissipation law [12, 13]. These ideas have been carried to finite-dimensional optimization [14, 15], including a relaxed vector auxiliary variable algorithm for unconstrained problems [16, 17, 18]. Standard SAV methods, however, primarily depend on the total gradient and therefore do not fully exploit the decomposition in Eq. (1).

The pullback-corrected SAV (PB–SAV) family for gradient flows [19] separates energy tracking from the curvature correction. A single scalar auxiliary variable tracks the total shifted objective, while the component gradients generate the positive semidefinite matrix

$$B_m = \sum_{i=1}^{m} \frac{\nabla E_i(\theta) \nabla E_i(\theta)^{\mathsf{T}}}{2\big(E_i(\theta) + C_i\big)}.$$

Here each shift C_i is chosen so that $E_i(\theta) + C_i > 0$. The matrix B_m retains the component gradient directions and has rank at most m. Its Gauss–Newton interpretation and the corresponding approximation of quadratic Hessians are established in [19]. In a direct Euclidean step, the aggregate SAV correction only rescales the gradient, while the component pullback can also change its direction. The correction therefore uses only component energies and first derivatives. Section 2 restates that construction. We build on it an optimizer with momentum and adaptive scaling, and determine how the component curvature enters its energy law and its local stability condition.

Our main contributions are as follows.

- **Component curvature and local stability.** Proposition 3.1 characterizes the Gauss–Newton curvature retained by a vector-valued least-squares component. For fixed mobility, Theorem 3.9 and Corollary 3.10 give the exact local stability condition and quantify the change in admissible step sizes relative to the aggregate correction, including the case in which the component gradients cancel, so that the aggregate correction vanishes while the component correction does not.

- **Momentum inside the solve.** The gradient and stored momentum enter one implicit equation. Proposition 3.4 identifies its quadratic model and the excess incurred by appending momentum after the solve, and Proposition 3.3 shows that the corresponding identity for the appended construction carries a cross term of indefinite sign. For the momentum update, Theorem 3.6 establishes an exact modified energy law for positive definite mobilities that are nonincreasing in the Loewner order, including Euclidean and AMSGrad-type choices.

- **Convergence of the relaxed update.** Under the local stability condition, Theorem 3.11 proves local geometric convergence near a nondegenerate minimum for fixed mobility and every scalar relaxation sequence. For varying mobility, Theorem 3.12 and Corollary 3.14 give stationarity and an $O(N^{-1/2})$ best-iterate gradient bound under uniform mobility and curvature bounds, a positive lower bound on q^n/Q^n, and relaxation parameters bounded away from one.

Section 2 introduces SAV and the PB–SAV geometry. Section 3 presents the method and analysis; the implementation in Algorithm 1 uses a dense solve of order at most m (Proposition 3.8). Section 4 reports five numerical experiments and Section 5 concludes. Longer proofs appear in Appendix A.

Appendix B gives a curvature condition for exact scalar tracking under full relaxation.

2. Preliminaries

2.1. Gradient flow and the scalar auxiliary variable

Let $F : \mathbb{R}^d \to \mathbb{R}$ be differentiable and bounded from below. Gradient descent is the forward Euler discretization of the Euclidean gradient flow $\dot{\theta} = -\nabla F(\theta)$, along which $\frac{\mathrm{d}}{\mathrm{d}t}F = -\left\|\nabla F\right\|^2 \leq 0$. Choose C with $F(\theta)+C > 0$ on the states of interest and set $Q(\theta) = \sqrt{F(\theta) + C}$ and $g(\theta) = \nabla F(\theta)$. In the SAV reformulation for gradient flows [10, 11], an auxiliary scalar q tracks $Q(\theta(t))$ and the modified energy q^2 satisfies $\frac{\mathrm{d}}{\mathrm{d}t}q^2 = -\left\|\dot{\theta}\right\|^2 \leq 0$. Convergence and error analysis for the resulting schemes [20] and a related energy-stable variant [21] are available.

For the discrete formulation, let $g^n = g(\theta^n)$ and $Q^n = Q(\theta^n)$, initialize $q^0 = Q(\theta^0)$, and let $\eta > 0$ be the step size. The first-order SAV update is

$$\frac{\theta^{n+1} - \theta^n}{\eta} = -\frac{q^{n+1}}{Q^n} g^n, \tag{2a}$$

$$q^{n+1} - q^n = \frac{(g^n)^{\mathsf{T}}(\theta^{n+1} - \theta^n)}{2Q^n}. \tag{2b}$$

Writing $\Delta\theta^{n+1} = \theta^{n+1} - \theta^n$ and eliminating q^{n+1} gives

$$\left(\eta^{-1} I_d + B_1^n\right) \Delta\theta^{n+1} = -\frac{q^n}{Q^n} g^n, \qquad B_1^n = \frac{g^n (g^n)^{\mathsf{T}}}{2(Q^n)^2}. \tag{3}$$

Here I_d is the $d \times d$ identity matrix, and identities of other sizes carry the corresponding subscript. The correction B_1^n has rank one, and $q^n/Q^n = 1$ when the stored scalar matches $Q(\theta^n)$.

2.2. PB–SAV for composite objectives

Consider an additive objective

$$F(\theta) = \sum_{i=1}^{m} E_i(\theta), \qquad E_i(\theta) + C_i > 0, \qquad C = \sum_{i=1}^{m} C_i. \tag{4}$$

Multiple-SAV methods assign a separate auxiliary variable to each energy component [22]. PB–SAV instead retains one auxiliary variable for the total shifted objective and builds the correction from the individual components [19].

At θ^n, define

$$Q_i^n = \sqrt{E_i(\theta^n) + C_i}, \qquad g_i^n = \nabla E_i(\theta^n), \qquad g^n = \sum_{i=1}^m g_i^n, \tag{5}$$

so that $(Q^n)^2 = \sum_{i=1}^m (Q_i^n)^2$. The component matrix is

$$V^n = \begin{bmatrix} \dfrac{g_1^n}{\sqrt{2}Q_1^n} & \cdots & \dfrac{g_m^n}{\sqrt{2}Q_m^n} \end{bmatrix}, \qquad B_m^n = V^n (V^n)^\mathsf{T}. \tag{6}$$

2.3. Component curvature

The PB–SAV pullback is a Gauss–Newton matrix for the transformed residuals [19]

$$\psi_i(\theta) = \sqrt{2(E_i(\theta) + C_i)}, \tag{7}$$

with $\psi = (\psi_1, \dots, \psi_m)^\mathsf{T}$ and $F + C = \frac{1}{2}\|\psi\|^2$. If $J_\psi(\theta)$ denotes its Jacobian, differentiation gives

$$B_m^n = J_\psi(\theta^n)^\mathsf{T} J_\psi(\theta^n). \tag{8}$$

For scalar least-squares components $E_i = \frac{1}{2} r_i^2$ with $C_i > 0$, this formula becomes

$$B_m = \sum_{i=1}^m \frac{r_i^2}{r_i^2 + 2C_i} \nabla r_i \nabla r_i^\mathsf{T}. \tag{9}$$

Thus the component pullback approximates the Gauss–Newton matrix $G = \sum_i \nabla r_i \nabla r_i^\mathsf{T}$ when the shifts are small relative to the squared residuals. For twice continuously differentiable residuals,

$$\nabla^2 F = G + \sum_i r_i \nabla^2 r_i, \tag{10}$$

so the residual-weighted second derivatives determine the difference between Gauss–Newton and Hessian curvature [23].

The relation to the aggregate correction follows from a weighted variance identity [19]. For any $z \in \mathbb{R}^d$, define

$$y_i^n = \frac{z^\mathsf{T} g_i^n}{(Q_i^n)^2}, \qquad \bar{y}^n = \frac{\sum_i (Q_i^n)^2 y_i^n}{\sum_i (Q_i^n)^2}. \tag{11}$$

Then

$$z^\mathsf{T}(B_m^n - B_1^n) z = \frac{1}{2} \sum_i (Q_i^n)^2 (y_i^n - \bar{y}^n)^2 \geq 0. \tag{12}$$

Thus $B_m^n - B_1^n \succeq 0$.

3. PB–SAV optimization with momentum and adaptive mobility

3.1. Component curvature and the direct PB–SAV step

Notation follows Section 2, with all component quantities evaluated at θ^n. Each E_i is continuously differentiable, and the shifted component energies are positive along the iterates. Fix $\eta > 0$ and initialize $q^0 = Q^0$. Define the curvature gap and the interpolation by

$$S_m^n = B_m^n - B_1^n, \qquad B_\alpha^n = B_1^n + \alpha S_m^n, \qquad 0 \le \alpha \le 1. \tag{13}$$

At $\alpha = 0$ this is the aggregate correction B_1^n, and at $\alpha = 1$ the component pullback B_m^n. The variance identity Eq. (12) gives $S_m^n \succeq 0$, so B_α^n increases with α in the Loewner order. We write $B_1(\theta)$, $B_m(\theta)$, and $B_\alpha(\theta)$ for the same matrices evaluated at an arbitrary state.

3.1.1. Curvature retained by a least-squares component

The scalar residual formula Eq. (9) extends to vector-valued residuals, which arise when a component contains many residual terms.

Proposition 3.1. *At a fixed parameter state, let $E_i = \frac{1}{2}\|r_i\|^2$, where $r_i : \mathbb{R}^d \to \mathbb{R}^{k_i}$ is continuously differentiable, and let $J_i = Dr_i$, $C_i > 0$, and $G = \sum_i J_i^{\mathsf{T}} J_i$. Then*

$$B_m = \sum_i J_i^{\mathsf{T}} \frac{r_i r_i^{\mathsf{T}}}{\|r_i\|^2 + 2C_i} J_i, \qquad 0 \preceq B_m \preceq G. \tag{14}$$

Define

$$\widetilde{r}_i = \begin{bmatrix} r_i \\ \sqrt{2C_i} \end{bmatrix}, \qquad \widetilde{J}_i = \begin{bmatrix} J_i \\ 0 \end{bmatrix}, \qquad u_i = \frac{\widetilde{r}_i}{\|\widetilde{r}_i\|}. \tag{15}$$

For every $v \in \mathbb{R}^d$,

$$v^{\mathsf{T}}(G - B_m)v = \sum_i \left\| (I_{k_i+1} - u_i u_i^{\mathsf{T}}) \widetilde{J}_i v \right\|^2. \tag{16}$$

Proof. Substitute $g_i = J_i^{\mathsf{T}} r_i$ into the definition of B_m. The matrix $u_i u_i^{\mathsf{T}}$ is an orthogonal projector, and $\widetilde{J}_i^{\mathsf{T}} \widetilde{J}_i = J_i^{\mathsf{T}} J_i$. Subtract the projected term from that identity and evaluate on v. □

Remark 3.1. Each least-squares component retains the part of its predicted residual change parallel to its current residual, carrying the weight $\|r_i\|^2 / (\|r_i\|^2 + 2C_i)$ of Eq. (14); the orthogonal part is discarded. At $r_i = 0$ the contribution vanishes even if $J_i^{\mathsf{T}} J_i \neq 0$. For a fixed nonzero residual, the weight tends to one as $C_i \to 0$ and decreases toward zero as C_i grows.

3.1.2. The direct update

For an increment $\Delta\theta^{n+1}$, define

$$d^{n+1} = \frac{(g^n)^{\mathsf{T}} \Delta\theta^{n+1}}{2Q^n}. \tag{17}$$

The scalar and pullback increments satisfy, with $\|x\|_A^2 = x^{\mathsf{T}} A x$ here and below (a seminorm when the symmetric positive semidefinite A is singular),

$$(q^n + d^{n+1})^2 - (q^n)^2 = 2q^n d^{n+1} + (d^{n+1})^2, \tag{18a}$$

$$\left\|\Delta\theta^{n+1}\right\|_{B_\alpha^n}^2 = 2(d^{n+1})^2 + \alpha \left\|\Delta\theta^{n+1}\right\|_{S_m^n}^2. \tag{18b}$$

The direct PB–SAV update [19] is

$$\left(\eta^{-1} I_d + B_\alpha^n\right) \Delta\theta^{n+1} = -\frac{q^n}{Q^n} g^n, \tag{19a}$$

$$\theta^{n+1} = \theta^n + \Delta\theta^{n+1}, \qquad q^{n+1} = q^n + d^{n+1}. \tag{19b}$$

Theorem 3.2. *The update Eq.* (19) *satisfies*

$$(q^n)^2 - (q^{n+1})^2 = \frac{\left\|\Delta\theta^{n+1}\right\|^2}{\eta} + (d^{n+1})^2 + \alpha \left\|\Delta\theta^{n+1}\right\|_{S_m^n}^2 \geq 0. \tag{20}$$

Proof. Take the inner product of Eq. (19a) with $\Delta\theta^{n+1}$. Substitute the two identities in Eq. (18), and move the change in scalar energy to the other side. Nonnegativity follows from $S_m^n \succeq 0$. □

3.2. Momentum placement and mobility

The direct update stores the scalar q^n but no previous increment. Momentum carries previous increments into the current step. Euclidean heavy-ball momentum [7] retains a fraction β of the previous parameter increment s^n,

$$s^{n+1} = \beta s^n - \eta g^n, \qquad \theta^{n+1} = \theta^n + s^{n+1}, \qquad 0 \leq \beta < 1, \tag{21}$$

and reduces to gradient descent at $\beta = 0$.

3.2.1. Placement of momentum

Consider first computing the direct PB–SAV increment and then appending heavy-ball memory:

$$\left(\eta^{-1} I_d + B_\alpha^n\right) \delta_{\mathrm{PB}}^{n+1} = -\frac{q^n}{Q^n} g^n, \tag{22a}$$

$$s^{n+1} = \delta_{\mathrm{PB}}^{n+1} + \beta s^n, \tag{22b}$$

$$\theta^{n+1} = \theta^n + s^{n+1}, \qquad q^{n+1} = q^n + d^{n+1}, \tag{22c}$$

where $d^{n+1} = (g^n)^\mathsf{T} s^{n+1}/(2Q^n)$. The scalar update uses the total increment. Define the Euclidean modified energy

$$\mathcal{H}^n_{\mathrm{app}} = (q^n)^2 + \frac{\|s^n\|^2}{2\eta}. \tag{23}$$

Proposition 3.3. *The update Eq.* (22) *satisfies*

$$\begin{aligned}\mathcal{H}^n_{\mathrm{app}} - \mathcal{H}^{n+1}_{\mathrm{app}} = \frac{\left\|s^{n+1} - \beta s^n\right\|^2}{2\eta} + \frac{1-\beta^2}{2\eta}\|s^n\|^2 \\ + (d^{n+1})^2 + \alpha \left\|s^{n+1}\right\|^2_{S^n_m} - \beta (s^{n+1})^\mathsf{T} B^n_\alpha s^n.\end{aligned} \tag{24}$$

The last term has no fixed sign.

Proof. The total increment obeys $(\eta^{-1} I_d + B^n_\alpha)(s^{n+1} - \beta s^n) = -(q^n/Q^n) g^n$. Take its inner product with s^{n+1} and apply Eq. (18). Subtracting the change in kinetic energy and using Euclidean polarization gives Eq. (24). □

3.2.2. The momentum and the mobility

We place momentum on the right-hand side of the implicit solve, so that the pullback acts on the gradient and memory together. This removes the extra $\beta B^n_\alpha s^n$ term in the appended state equation. Let $p^n \in \mathbb{R}^d$ be the momentum and let the mobility $M^n \in \mathbb{R}^{d\times d}$, symmetric positive definite, map it to the parameter increment $M^n p^n$; this is the mobility in gradient-flow terminology and a variable-metric preconditioner in optimization. The new momentum and increment are related by

$$\Delta\theta^{n+1} = M^{n+1} p^{n+1}, \qquad p^{n+1} = (M^{n+1})^{-1} \Delta\theta^{n+1}. \tag{25}$$

We assume that the mobility is nonincreasing in the Loewner order:

$$0 \prec M^{n+1} \preceq M^n. \tag{26}$$

The PB–SAV momentum update is

$$\left((M^{n+1})^{-1} + \eta B^n_\alpha\right) \Delta\theta^{n+1} = \beta p^n - \eta \frac{q^n}{Q^n} g^n, \tag{27}$$

or, equivalently,

$$\frac{p^{n+1} - \beta p^n}{\eta} + B^n_\alpha \Delta\theta^{n+1} + \frac{q^n}{Q^n} g^n = 0. \tag{28}$$

With Euclidean mobility, omitting the pullback term and setting $q^n/Q^n = 1$ in this balance gives the heavy-ball recursion Eq. (21), with $p^n = s^n$. The parameter and scalar states remain

$$\theta^{n+1} = \theta^n + \Delta\theta^{n+1}, \qquad q^{n+1} = q^n + d^{n+1}. \tag{29}$$

3.2.3. The inertial quadratic model

The implicit solve Eq. (27) has a quadratic optimization interpretation.

Proposition 3.4. *The increment $\Delta\theta^{n+1}$ solving Eq. (27) is the unique minimizer of*

$$\mathcal{Q}(s) = \frac{q^n}{Q^n}(g^n)^\mathsf{T} s + \frac{1}{2} s^\mathsf{T} B_\alpha^n s + \frac{1}{2\eta} \left\| s - \beta M^{n+1} p^n \right\|^2_{(M^{n+1})^{-1}}. \tag{30}$$

Appending the momentum after the solve gives instead

$$\Delta_{\text{app}}^{n+1} = -\eta \frac{q^n}{Q^n} \big((M^{n+1})^{-1} + \eta B_\alpha^n \big)^{-1} g^n + \beta M^{n+1} p^n, \tag{31}$$

and the two increments satisfy

$$\Delta_{\text{app}}^{n+1} - \Delta\theta^{n+1} = \beta\eta \big((M^{n+1})^{-1} + \eta B_\alpha^n \big)^{-1} B_\alpha^n M^{n+1} p^n, \tag{32}$$

$$\mathcal{Q}(\Delta_{\text{app}}^{n+1}) - \mathcal{Q}(\Delta\theta^{n+1}) = \frac{1}{2\eta} \left\| \Delta_{\text{app}}^{n+1} - \Delta\theta^{n+1} \right\|^2_{(M^{n+1})^{-1} + \eta B_\alpha^n} \geq 0. \tag{33}$$

Proof. The Hessian of $\mathcal{Q}$ is $B_\alpha^n + \eta^{-1}(M^{n+1})^{-1} \succ 0$, and its optimality equation is Eq. (27). Subtracting the two increments and using $\big((M^{n+1})^{-1} + \eta B_\alpha^n\big) M^{n+1} = I_d + \eta B_\alpha^n M^{n+1}$ gives Eq. (32). Expanding $\mathcal{Q}$ about its minimizer gives Eq. (33). □

Remark 3.2. For a fixed α, let $\Psi_\alpha(\theta)$ stack $\sqrt{2(1-\alpha)}\, Q(\theta)$ on $\sqrt{\alpha}\, \psi(\theta)$, so that $\frac{1}{2}\|\Psi_\alpha\|^2 = F + C$, its Gauss–Newton matrix is B_α, and its gradient is g. At exact tracking $q^n = Q^n$, the shifted model $\mathcal{Q}(s) + F + C$ is a Gauss–Newton model for the residual map Ψ_α, together with a proximal term centered at $\beta M^{n+1} p^n$ and measured in $\|\cdot\|_{(M^{n+1})^{-1}}$; otherwise the linear term is scaled by q^n/Q^n. Levenberg–Marquardt methods for neural networks also combine curvature with adaptive momentum [24], without the scalar coupling that produces the energy law below.

To see how the solve acts on memory, let $b_j \geq 0$ be the eigenvalues of $(M^{n+1})^{1/2} B_\alpha^n (M^{n+1})^{1/2}$ and let u_j be the coordinates of $(M^{n+1})^{1/2} p^n$ in a corresponding orthonormal eigenbasis. In those coordinates the solve multiplies the jth component of the stored momentum by $\beta/(1+\eta b_j)$, and

$$\mathcal{Q}(\Delta_{\text{app}}^{n+1}) - \mathcal{Q}(\Delta\theta^{n+1}) = \frac{\beta^2 \eta}{2} \sum_j \frac{b_j^2}{1+\eta b_j} u_j^2. \tag{34}$$

Remark 3.3. Directions with larger b_j damp the stored momentum more strongly, and where $b_j = 0$ the coefficient remains β. The excess in Eq. (34) is measured in the quadratic model, not in the modified energy of Theorem 3.6.

3.2.4. Curvature action of the momentum step

The following comparison measures the approximation error of the correction through its action on the reference increment.

Proposition 3.5. *Let $M \succ 0$, let $A, B \succeq 0$ be symmetric, and let $\eta > 0$. For a common right-hand side $f \in \mathbb{R}^d$, define*

$$(M^{-1} + \eta B)\Delta_B = f, \qquad (M^{-1} + \eta A)\Delta_A = f. \tag{35}$$

Then

$$\Delta_B - \Delta_A = \eta(M^{-1} + \eta B)^{-1}(A - B)\Delta_A, \tag{36}$$

and

$$\|\Delta_B - \Delta_A\|_{M^{-1}} \le \eta \left\| M^{1/2}(A - B)\Delta_A \right\|. \tag{37}$$

In particular, $\Delta_B = \Delta_A$ if and only if $(A - B)\Delta_A = 0$.

Proof. Subtracting the two systems gives Eq. (36). Since $M^{1/2}BM^{1/2} \succeq 0$, we have $\left\|(I_d + \eta M^{1/2}BM^{1/2})^{-1}\right\|_2 \le 1$; multiplying Eq. (36) by $M^{-1/2}$ then gives Eq. (37). The equality criterion follows from invertibility of $M^{-1} + \eta B$. □

Remark 3.4. For the momentum update, take $M = M^{n+1}$, $B = B_\alpha^n$, and $f = \beta p^n - \eta(q^n/Q^n)g^n$; a Gauss–Newton matrix, or a positive semidefinite Hessian at the current state, can serve as A. A low-rank B approximates Δ_A when the weighted action error on the right-hand side of Eq. (37) is small, even when $A - B$ is not small in operator norm, and reproduces it exactly when $(A - B)\Delta_A = 0$. The quadratic and regression tests in Sections 4.1 and 4.2 measure the matrix error and the step error separately.

3.3. Dissipation and mobility choices

3.3.1. Modified energy law

Define, for every n,

$$\mathcal{H}^n_{\mathrm{mom}} = (q^n)^2 + \frac{1}{2\eta}\,\|p^n\|^2_{M^n}. \tag{38}$$

Theorem 3.6. *For $0 \le \beta < 1$ and $0 \prec M^{n+1} \preceq M^n$, the update Eqs. (25), (27) and (29) satisfies*

$$\begin{aligned}\mathcal{H}^n_{\mathrm{mom}} - \mathcal{H}^{n+1}_{\mathrm{mom}} = {} & \frac{\|p^n\|^2_{M^n - M^{n+1}}}{2\eta} + \frac{\left\|p^{n+1} - \beta p^n\right\|^2_{M^{n+1}}}{2\eta} \\ & + \frac{1-\beta^2}{2\eta}\,\|p^n\|^2_{M^{n+1}} + (d^{n+1})^2 + \alpha\left\|\Delta\theta^{n+1}\right\|^2_{S^n_m} \ge 0.\end{aligned} \tag{39}$$

The proof is a weighted polarization of Eq. (28) against $\Delta\theta^{n+1} = M^{n+1}p^{n+1}$, followed by substitution of Eq. (18); it is given in Appendix A.1.

Remark 3.5. The dissipated quantity is the modified energy $\mathcal{H}^n_{\mathrm{mom}}$, which controls the scalar and kinetic energies together rather than the objective $F(\theta^n)$ alone. Its five nonnegative terms are, in order, the dissipation from the mobility change, the inertial residual, the momentum damping, the scalar tracking term, and the curvature gap, which the relaxation step of Section 3.4 sums.

3.3.2. Euclidean momentum

Set $M^n = I_d$ and identify $p^n = s^n$. The abstract state equation becomes

$$(I_d + \eta B^n_\alpha)\, s^{n+1} = \beta s^n - \eta \frac{q^n}{Q^n} g^n. \tag{40}$$

The constant mobility $M^n \equiv I_d$ satisfies Eq. (26) trivially, so Theorem 3.6 applies for every $0 \le \beta < 1$; its mobility-change term vanishes, and the modified energy is $(q^n)^2 + \|s^n\|^2/(2\eta)$.

3.3.3. AMSGrad-type diagonal mobility

An AMSGrad-type maximum envelope [9] gives a diagonal mobility that is nonincreasing in the Loewner order. Let $0 \le \beta_2 < 1$, $\epsilon > 0$, and initialize $v^0 = \overline{v}^0 = 0$. With $\odot$ the componentwise product and the square root, division, and maximum below taken componentwise, define

$$v^{n+1} = \beta_2 v^n + (1-\beta_2)(g^n \odot g^n), \tag{41a}$$

$$\widehat{v}^{n+1} = \frac{v^{n+1}}{1-\beta_2^{n+1}}, \tag{41b}$$

$$\overline{v}^{n+1} = \max\{\overline{v}^n, \widehat{v}^{n+1}\}, \tag{41c}$$

$$M^{n+1} = \operatorname{diag}\left(\frac{1}{\sqrt{\overline{v}^{n+1}} + \epsilon}\right). \tag{41d}$$

Then $M^0 = \epsilon^{-1} I_d$, and the maximum envelope implies

$$0 \prec M^{n+1} \preceq M^n, \tag{42}$$

so Theorem 3.6 applies to this mobility for every $0 \le \beta < 1$.

3.4. Scalar relaxation

Relaxed-SAV schemes modify the auxiliary state while preserving a discrete energy bound [12, 13, 25]. The squared-scalar update below is the

relaxation of [19]; it depends on the update only through its Lyapunov functional, so one statement covers both updates of this section.

Without relaxation, $q^{n+1} = q^n + d^{n+1}$ is the stored scalar in Eqs. (19) and (29). When relaxation is applied, the same algebraic output is provisional and is denoted by

$$\bar{q}^{n+1} = q^n + d^{n+1}. \tag{43}$$

For the direct update the old and provisional energies are $\mathcal{H}^n_{\mathrm{dir}} = (q^n)^2$ and $\bar{\mathcal{H}}^{n+1}_{\mathrm{dir}} = (\bar{q}^{n+1})^2$; for the momentum update they are $\mathcal{H}^n_{\mathrm{mom}} = (q^n)^2 + \|p^n\|^2_{M^n}/(2\eta)$ and $\bar{\mathcal{H}}^{n+1}_{\mathrm{mom}} = (\bar{q}^{n+1})^2 + \left\|p^{n+1}\right\|^2_{M^{n+1}}/(2\eta)$.

Proposition 3.7. *For either update, let $(\mathcal{H}^n, \bar{\mathcal{H}}^{n+1})$ denote the corresponding pair above and suppose that*

$$\mathcal{H}^n - \bar{\mathcal{H}}^{n+1} = \mathcal{D}^n \geq 0. \tag{44}$$

For $0 \leq \rho^n \leq 1$, define

$$(q^{n+1})^2 = \min\left\{Q(\theta^{n+1})^2,\ (\bar{q}^{n+1})^2 + \rho^n \mathcal{D}^n\right\}, \qquad q^{n+1} \geq 0. \tag{45}$$

Let $\mathcal{H}^{n+1}$ be the same energy evaluated with the accepted scalar q^{n+1}. Then

$$\mathcal{H}^{n+1} \leq \mathcal{H}^n - (1-\rho^n)\mathcal{D}^n \leq \mathcal{H}^n. \tag{46}$$

Proof. Relaxation changes only the scalar part of the new energy. Therefore $\mathcal{H}^{n+1} \leq \bar{\mathcal{H}}^{n+1} + \rho^n \mathcal{D}^n$, and Eq. (44) gives Eq. (46). □

With $q^0 = Q(\theta^0)$, the accepted scalars satisfy $0 \leq q^n \leq Q(\theta^n)$ by Eq. (45). Full relaxation $\rho^n = 1$ preserves the modified energy bound; choosing $\rho^n \leq \bar{\rho} < 1$ also gives $\sum_n \mathcal{D}^n \leq \mathcal{H}^0/(1-\bar{\rho})$. Under the curvature condition in Appendix B, full relaxation preserves $q^n = Q(\theta^n)$, so the momentum energy becomes $F(\theta^n) + C + \|p^n\|^2_{M^n}/(2\eta)$.

3.5. Low-rank implementation and cost per iteration

With V^n as in Eq. (6) and $c^n = \frac{1}{Q^n}[\, Q^n_1 \ \cdots \ Q^n_m \,]^{\mathsf{T}}$, we have $\|c^n\| = 1$ and

$$B^n_\alpha = V^n \left[\alpha I_m + (1-\alpha) c^n (c^n)^{\mathsf{T}}\right] (V^n)^{\mathsf{T}}. \tag{47}$$

Consequently $\mathrm{rank}(B^n_\alpha) \leq \mathrm{rank}(V^n) \leq m$. For a concrete factor with at most m columns, define

$$R^n = \left[\alpha I_m + (1-\alpha) c^n (c^n)^{\mathsf{T}}\right]^{1/2}, \qquad W^n = V^n R^n. \tag{48}$$

Because $\|c^n\| = 1$, the symmetric positive semidefinite square root has the closed form $R^n = \sqrt{\alpha}\, I_m + (1 - \sqrt{\alpha}) c^n (c^n)^{\mathsf{T}}$, which at $\alpha = 0$ is the rank-one

projector $c^n(c^n)^{\mathsf{T}}$. It also satisfies $W^n(W^n)^{\mathsf{T}} = B_\alpha^n$ with $\operatorname{rank}(W^n) \le m$. With

$$f^n = \beta p^n - \eta \frac{q^n}{Q^n} g^n, \tag{49}$$

the momentum equation is

$$\left((M^{n+1})^{-1} + \eta W^n (W^n)^{\mathsf{T}} \right) \Delta\theta^{n+1} = f^n. \tag{50}$$

Algorithm 1 Relaxed PB–SAV momentum with nonincreasing mobility. Use $M^n \equiv I_d$ for Euclidean momentum; for the AMSGrad-type mobility use $v^0 = \overline{v}^0 = 0$ and $M^0 = \epsilon^{-1} I_d$.

Require: Components $\{E_i, C_i\}_{i=1}^m$, θ^0, $q^0 = Q(\theta^0)$, $p^0 = 0$, number of updates N, $\eta > 0$, $\beta \in [0,1)$, $\alpha \in [0,1]$, relaxation schedule $\{\rho^n\}_{n=0}^{N-1} \subset [0,1]$, and a mobility initialization and update rule satisfying $0 \prec M^{n+1} \preceq M^n$

1: **for** $n = 0, \dots, N-1$ **do**
2: Evaluate $E_i(\theta^n)$, g_i^n, Q_i^n, Q^n, and $g^n = \sum_i g_i^n$.
3: Update M^{n+1} using the prescribed nonincreasing mobility rule.
4: Form W^n from Eq. (48).
5: Form f^n from Eq. (49).
6: Compute $\Delta\theta^{n+1}$ from Eq. (51).
7: Set $p^{n+1} = (M^{n+1})^{-1} \Delta\theta^{n+1}$ and $\theta^{n+1} = \theta^n + \Delta\theta^{n+1}$.
8: Compute $\bar{q}^{n+1}$ from Eq. (43).
9: Sum the five terms in Eq. (39) as $\mathcal{D}^n$.
10: Re-evaluate $Q(\theta^{n+1})$ and apply Eq. (45).
11: **end for**

Ensure: θ^N

Proposition 3.8. *The unique solution of Eq.* (50) *is*

$$\begin{aligned} \Delta\theta^{n+1} = M^{n+1} f^n - \eta M^{n+1} W^n \left(I_m + \eta (W^n)^{\mathsf{T}} M^{n+1} W^n \right)^{-1} \\ \times (W^n)^{\mathsf{T}} M^{n+1} f^n. \end{aligned} \tag{51}$$

Besides applying the mobility, the computation requires a dense system of order at most m.

Proof. The coefficient in Eq. (50) is positive definite because $M^{n+1} \succ 0$ and $W^n(W^n)^{\mathsf{T}} \succeq 0$. The Sherman–Morrison–Woodbury identity [26, 27] with base inverse M^{n+1} gives Eq. (51). □

Remark 3.6. Each iteration evaluates the m component energies and gradients that the algorithm already forms. The factor R^n is available in closed form, so

$$W^n = \sqrt{\alpha}\, V^n + \bigl(1 - \sqrt{\alpha}\bigr)\bigl(V^n c^n\bigr)(c^n)^{\mathsf{T}}$$

costs one matrix–vector product and needs no eigensolver. Evaluating Eq. (51) requires the Gram matrix $(W^n)^{\mathsf{T}} M^{n+1} W^n$ and one dense solve of order at most m, so the work beyond applying the mobility is $O(m^2 d + m^3)$. Storage beyond the iterate is the $d \times m$ factor W^n, the momentum p^n, and the mobility state. The relaxation step re-evaluates $Q(\theta^{n+1})$, one objective evaluation per iteration beyond the gradient.

The direct update omits the momentum and mobility states: it solves $\bigl(I_d + \eta W^n (W^n)^{\mathsf{T}}\bigr)\,\Delta\theta^{n+1} = f^n_{\mathrm{dir}}$ with $f^n_{\mathrm{dir}} = -\eta \frac{q^n}{Q^n} g^n$, using the same reduction to a dense system of order at most m, and takes $\mathcal{D}^n$ in the scalar relaxation from Eq. (20).

3.6. Local stability and convergence

3.6.1. Local stability and retained curvature

With fixed mobility, the pullback changes the local stability condition at a stationary point θ_* of F. Throughout this subsection a subscript $*$ denotes evaluation at that point, so that $H_* = \nabla^2 F(\theta_*)$ and $B_* = B_\alpha(\theta_*)$, and (θ_*, p_*, q_*) is the corresponding state of Eqs. (27) and (29). The comparison depends on the Hessian curvature left after subtracting twice B_*.

Theorem 3.9. *Let θ_* satisfy $\nabla F(\theta_*) = 0$, suppose the component energies are twice continuously differentiable near θ_* with positive shifted energies, and let $H_* \succ 0$. Fix $0 \le \alpha \le 1$, $M \succ 0$, $\eta > 0$, and $0 \le \beta < 1$. The corresponding rest state has $p_* = 0$ and a stored scalar $q_* > 0$; set $\kappa_* = q_*/Q(\theta_*)$. The parameter–momentum block of the linearization of Eqs.* (27) *and* (29) *is Schur stable, meaning that all its eigenvalues have modulus less than one, if and only if*

$$\kappa_* H_* \prec 2B_* + \frac{2(1+\beta)}{\eta} M^{-1}. \tag{52}$$

The proof, in Appendix A.2, reduces the linearized parameter–momentum map to a matrix polynomial and then to a scalar quadratic in the Rayleigh quotients $b = \xi^* \widehat{B}_* \xi / (\xi^* \xi)$ and $h = \xi^* \widehat{H}_* \xi / (\xi^* \xi)$, where $\widehat{B}_* = M^{1/2} B_* M^{1/2}$, $\widehat{H}_* = M^{1/2} H_* M^{1/2}$, and ξ^* is the conjugate transpose of $\xi \in \mathbb{C}^d$.

The unrelaxed scalar equation has a neutral direction at the stationary state; the scalar is controlled separately in the local convergence argument below.

Corollary 3.10. *Under the assumptions of Theorem 3.9, define*

$$L_{\mathrm{eff}} = \lambda_{\max}\Big(M^{1/2}(\kappa_* H_* - 2B_*)M^{1/2}\Big). \tag{53}$$

If $L_{\mathrm{eff}} > 0$, the exact stability interval is

$$0 < \eta < \frac{2(1+\beta)}{L_{\mathrm{eff}}}. \tag{54}$$

If $L_{\mathrm{eff}} \le 0$, every finite $\eta > 0$ is admissible. At the stationary point,

$$B_1(\theta_*) = 0, \qquad B_* = \alpha B_m(\theta_*), \qquad \operatorname{rank} B_m(\theta_*) \le \min\{d, m-1\}. \tag{55}$$

For a common θ_*, q_*, M, and β, increasing α cannot shrink the stability interval relative to the aggregate choice $\alpha = 0$.

Proof. Apply a congruence by $M^{1/2}$ to Eq. (52). The total gradient vanishes at θ_*, giving $B_1(\theta_*) = 0$. The component gradients sum to zero and are therefore linearly dependent, which gives the rank bound. Finally, $B_m \succeq 0$ makes L_{eff} nonincreasing with α. □

Remark 3.7. Gradient cancellation can leave nonzero component curvature after the aggregate correction has vanished; if all component gradients vanish, so does the component correction. Since $H_* \succ 0$, stability for every $\eta > 0$ requires $B_* \succ 0$, which Eq. (55) excludes once $m - 1 < d$; a low-rank correction can nevertheless enlarge a finite interval. The comparison in Corollary 3.10 is between component and aggregate pullbacks within the momentum update; Appendix A.3 records the corresponding comparison with the frozen appended recurrence, whose stability interval is at least as large.

3.6.2. Local convergence under any scalar relaxation

Theorem 3.11. *Let the component energies be twice continuously differentiable near θ_*, with positive shifted energies, $\nabla F(\theta_*) = 0$, and $H_* \succ 0$. Fix $M \succ 0$, $\eta > 0$, $0 \le \beta < 1$, and $0 \le \alpha \le 1$, and assume*

$$H_* \prec 2B_\alpha(\theta_*) + \frac{2(1+\beta)}{\eta} M^{-1}. \tag{56}$$

Use the momentum update with $M^n \equiv M$, the relaxation Eq. (45) with any sequence $\rho^n \in [0,1]$, and $q^0 = Q(\theta^0)$. There exist $\delta > 0$, $c_0, c_1 < \infty$, and $r \in (0,1)$ such that, whenever $a_0 = \|\theta^0 - \theta_*\| + \|p^0\| < \delta$, the iterates remain near θ_* and

$$\|\theta^n - \theta_*\| + \|p^n\| \le c_0 r^n a_0. \tag{57}$$

Moreover, $q^n \to q_\infty > 0$*, with*

$$0 \le Q(\theta_*) - q_\infty \le c_1 a_0^2. \tag{58}$$

The constants are uniform over all such relaxation sequences, including $\rho^n \equiv 1$*.*

The proof in Appendix A.4 combines contraction of the parameter–momentum map with a quadratic bound on the scalar drift.

3.7. Conditional stationarity

For a varying mobility, a global stationarity criterion follows from relaxation parameters bounded away from one and from a ratio q^n/Q^n bounded below by a positive constant. Assume

$$0 \le \rho^n \le \bar{\rho} < 1. \tag{59}$$

The proofs of the following estimates are given in Appendix A.5.

Theorem 3.12. *Consider the relaxed momentum update Eqs.* (25)*,* (27) *and* (45) *with a fixed* $0 \le \beta < 1$*. Suppose there exist constants*

$$0 < \mu_{\min} \le \mu_{\max} < \infty, \qquad \underline{\kappa} > 0, \tag{60}$$

such that, for every n*,*

$$\mu_{\min} I_d \preceq M^n \preceq \mu_{\max} I_d, \qquad \frac{q^n}{Q^n} \ge \underline{\kappa}, \qquad 0 \prec M^{n+1} \preceq M^n. \tag{61}$$

Assume also that $\sup_n \|B_\alpha^n\| < \infty$ *and Eq.* (59) *holds. Then*

$$p^n \to 0, \qquad p^{n+1} - \beta p^n \to 0, \qquad \Delta\theta^{n+1} \to 0, \qquad g^n \to 0. \tag{62}$$

Consequently, every accumulation point of $\{\theta^n\}$ *is stationary.*

Remark 3.8. If $E_i \ge 0$, $C_i > 0$, and $\sup_n \|g_i^n\| \le \Gamma_i$, then

$$\|B_\alpha^n\| \le \|B_m^n\| \le \sum_{i=1}^m \frac{\Gamma_i^2}{2C_i}. \tag{63}$$

For the diagonal realization Eq. (41d), $\sup_n \|g^n\|_\infty \le \Gamma_\infty$ gives

$$\frac{1}{\Gamma_\infty + \epsilon} I_d \preceq M^n \preceq \frac{1}{\epsilon} I_d. \tag{64}$$

The lower bound on q^n/Q^n must still be assumed separately.

Corollary 3.13. *Consider the relaxed direct update Eqs.* (19) *and* (45)*. Suppose Eq.* (59) *holds,* $\sup_n \|B_\alpha^n\| < \infty$*, and there exists* $\underline{\kappa} > 0$ *such that* $q^n/Q^n \ge \underline{\kappa}$ *for every* n*. Then* $\Delta\theta^{n+1} \to 0$*,* $g^n \to 0$*, and every accumulation point is stationary.*

Corollary 3.14. *Let $L_B = \sup_n \|B_\alpha^n\|$. Under the assumptions of Theorem 3.12 or of Corollary 3.13, there is a constant K, given explicitly in Appendix A.5, depending for the momentum update only on η, β, $\underline{\kappa}$, the mobility bounds, and L_B, and for the direct update only on η, $\underline{\kappa}$, and L_B. For either update, let $\mathcal{H}^0$ be its initial modified energy. Then, for every $N \geq 1$,*

$$\sum_{n=0}^{N-1} \|g^n\|^2 \leq \frac{K\mathcal{H}^0}{1-\bar{\rho}}, \qquad \min_{0\leq n<N} \|g^n\|^2 \leq \frac{K\mathcal{H}^0}{(1-\bar{\rho})N}. \tag{65}$$

Consequently, $\sum_{n=0}^{\infty} \|g^n\|^2 < \infty$, and the best gradient norm among the first N iterates is $O(N^{-1/2})$.

Full relaxation admits a different sufficient condition: the pullback must cover the true objective remainder along each step, up to a fixed fraction of the inertial dissipation. Under this condition, and under the same uniform mobility and pullback bounds as Theorem 3.12, Theorem B.2 gives exact tracking and stationarity without a separate lower bound on q^n/Q^n.

4. Numerical experiments

The experiments examine two aspects of component curvature: how accurately the component pullback reproduces Newton or Gauss–Newton increments, and whether resolving the objective into components improves practical optimization. The quadratic and regression tests compare curvature matrices and the increments they produce at shared states, as in Proposition 3.5. The forward Burgers experiment compares one and four components with matched learning rates and momentum settings. Darcy operator learning and inverse Burgers identification assess the complete optimizer of Algorithm 1 with its selected learning rates.

All calculations use double precision and deterministic full batches. The neural-network experiments were performed with PyTorch 2.5.1 and CUDA 12.4 on an NVIDIA A100 80GB GPU. Within each paired comparison, all methods start from the same parameters and use the same training, validation, and test data. Hyperparameters and model choices use only the training or validation quantities specified below; test errors are reported after selection. Curves show all recorded iterates without smoothing. The representative Darcy field is selected by a criterion independent of optimizer output; the inverse Burgers panel uses the median PB–SAV seed by final viscosity error, so that no best-case run is displayed. The Darcy and Burgers results use five paired seeds and report the seed mean with a band of one sample standard deviation.

Heavy-ball gradient descent applies the weight-decay coefficient as coupled L^2 regularization and AdamW applies it as decoupled weight decay [28]; because these mechanisms are not identical, each comparison reports the objective that is common to all methods.

4.1. Quadratic problem: Newton increments without full Hessian recovery

We use the quadratic benchmark with two curvature classes from [19] and add a comparison of full Hessian error with the error in the damped Newton increment:

$$F(\phi) = \sum_{j=1}^{100} a_j \phi_j^2, \qquad a_{2k-1} = 1, \qquad a_{2k} = 10^{-2}. \tag{66}$$

Its Hessian is $H = 2\,\mathrm{diag}(a_1, \dots, a_{100})$, and the minimizer is $\phi_* = 0$. The two distinct values of a_j form two curvature classes. We test whether a low-rank component split can reproduce the damped Newton increment even when it does not recover the full matrix H.

All methods start from $\phi^0 = \mathbf{1} \in \mathbb{R}^{100}$ and run for 300 updates. PB–SAV uses $\eta = 10$, and the shift associated with each coordinate energy is 10^{-12}. The one-component formulation aggregates all coordinates. The two-component formulation separates the two curvature classes. The 100-component formulation assigns one energy component to each coordinate. The intermediate component counts $4, 8, 16, 32, 64$ form a nested refinement and are used in the fixed-state geometry comparison. All PB–SAV runs in this subsection use the direct update with $\alpha = 1$.

Gradient descent uses the analytically optimal constant rate $2/(\lambda_{\min}(H) + \lambda_{\max}(H)) = 0.990099$. The reference direction is the exact damped Newton increment

$$\Delta_{\mathrm{DN}} = -(\eta^{-1} I_d + H)^{-1} \nabla F(\phi). \tag{67}$$

The matrix error is measured in the relative Frobenius norm, and the step error is measured relative to Δ_{DN} in the Euclidean norm. Both are evaluated at the shared initial state.

The fixed-state comparison in Fig. 2 separates the error in the matrix from the error in the increment it produces. The two-component correction has a relative Hessian error of 9.899×10^{-1}, but its step error is only 3.022×10^{-12}. The 100-component correction recovers the full Hessian to 1.414×10^{-12}, yet it produces the same step accuracy as the two-component formulation. The trajectories in Fig. 1 and Table 1 follow the same pattern. The two PB–SAV variants reach the 10^{-10} target in 63 updates, compared with 62 updates for exact damped Newton. One-component SAV converges more slowly, and

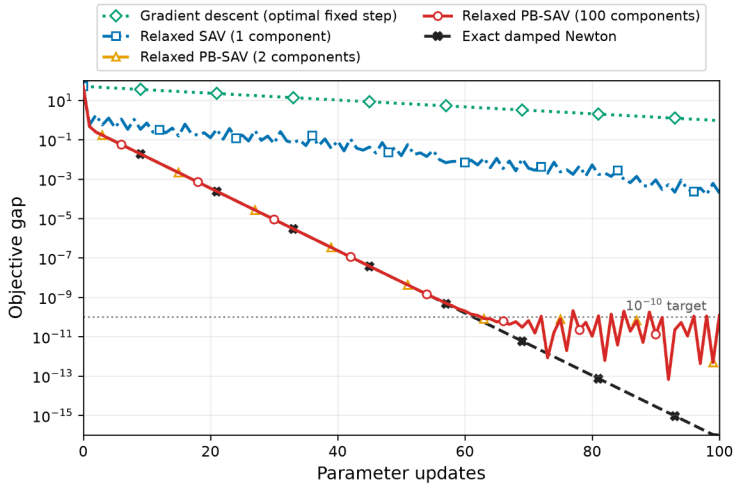

Figure 1: Convergence on the quadratic problem during the first 100 updates. All methods start from the same state.

optimally tuned fixed-step gradient descent remains above the target after 300 updates.

At ϕ^0, the gradient lies in the two-dimensional subspace spanned by the indicator vectors of the two curvature classes. The two-component pullback agrees with the damped Hessian on this subspace, but not outside it.

4.2. Nonlinear regression: Gauss–Newton recovery

The second example tests the pullback interpretation in a nonlinear least-squares problem. We fit $N = 30$ noisy observations of $f_*(x) = \sin(2\pi x)$ for $x \in [0, 1]$, where the observation noise is independent Gaussian noise with standard deviation 0.1. The model is a one-hidden-layer tanh network,

$$f_\theta(x) = \sum_{j=1}^{50} c_j \tanh(a_j x + b_j) + b_0, \tag{68}$$

with 151 trainable parameters. The model is therefore overparameterized relative to the 30 observations. The objective is

$$F(\theta) = \frac{1}{2N} \sum_{i=1}^{N} \left(f_\theta(x_i) - y_i\right)^2 . \tag{69}$$

For a pointwise least-squares decomposition, Eq. (9) relates the PB–SAV pullback to the Gauss–Newton matrix, with scaled residuals $r_i = (f_\theta(x_i) -$

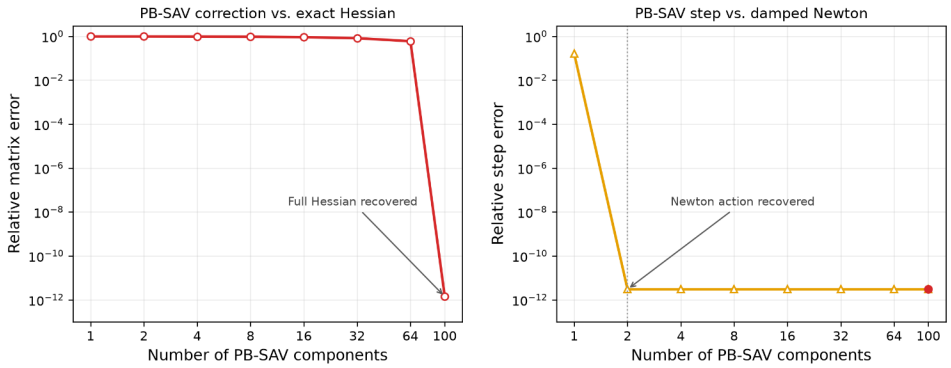

Figure 2: Fixed-state geometry for the quadratic problem. Left: relative error between the PB–SAV correction and the exact Hessian. Right: relative error between the PB–SAV increment and the damped Newton increment.

Table 1: Quadratic comparison after 300 updates. The Hessian and step errors are evaluated at the common initial state. The final gap is $F(\phi^{300}) - F(\phi_*)$, with $F(\phi_*) = 0$. A dash indicates that the quantity is not applicable or that the target was not reached within 300 updates. The columns report different quantities and are not ranked, so no entry is bold.

Method	Hessian error	Newton-step error	Updates to 10^{-10}	Final gap
Gradient descent (optimal step)	–	–	–	3.102×10^{-4}
Relaxed SAV (1 comp.)	9.900×10^{-1}	1.627×10^{-1}	–	1.873×10^{-10}
Relaxed PB–SAV (2 comp.)	9.899×10^{-1}	3.022×10^{-12}	63	7.879×10^{-11}
Relaxed PB–SAV (100 comp.)	1.414×10^{-12}	3.022×10^{-12}	63	9.623×10^{-11}
Exact damped Newton	–	0	62	1.550×10^{-48}

$y_i)/\sqrt{N}$. This example tests the relation at both the matrix and trajectory levels.

Nested random partitions use $1, 2, 4, 8, 16$, and 30 components. The final partition assigns one squared residual to each observation. PB–SAV and damped Gauss–Newton use the same adaptive damping controller, with $10^{-6} \leq \eta \leq 100$, and the same Armijo acceptance test [29] on the true objective. Gradient descent uses the rate 0.22, selected by a paired 200-update training-objective pilot. Adam is piloted over 10^{-3}, 3×10^{-3}, 10^{-2}, and 3×10^{-2}, and selects 3×10^{-2}. Each reported run contains 600 accepted updates. Prediction is measured by the mean-squared error (MSE) against the noise-free target. The PB–SAV runs use the direct update with $\alpha = 1$.

Figure 3 shows the trajectories and Fig. 4 the fixed-state errors. At the common initial state, the pointwise PB–SAV correction differs from the

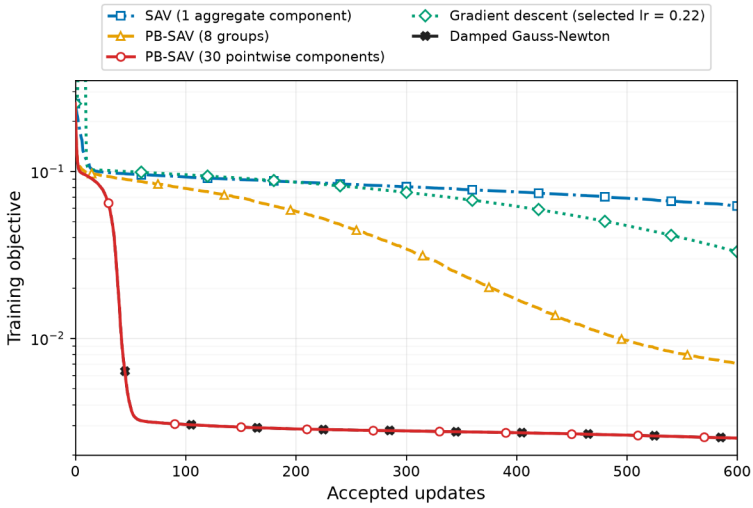

Figure 3: Training objective for one, eight, and 30 PB–SAV components, together with gradient descent and damped Gauss–Newton. The pointwise PB–SAV and Gauss–Newton trajectories are indistinguishable at plotting resolution.

Table 2: Nonlinear-regression results after 600 accepted updates. No entry is bold: PB–SAV and damped Gauss–Newton agree in every column to the displayed precision.

Method	Updates	Final objective	Best noise-free test MSE	Final $\|\nabla F\|$
PB–SAV (30 pointwise comp.)	600	2.525×10^{-3}	2.804×10^{-3}	1.032×10^{-4}
Damped Gauss–Newton	600	2.525×10^{-3}	2.804×10^{-3}	1.032×10^{-4}
Gradient descent (rate 0.22)	600	3.301×10^{-2}	6.040×10^{-2}	2.468×10^{-2}
Adam	600	2.985×10^{-3}	3.113×10^{-3}	6.695×10^{-4}

Gauss–Newton matrix by only 5.82×10^{-10}. The relative step errors for $\eta = 0.1, 1, 10, 100$ are 2.31×10^{-10}, 3.71×10^{-10}, 9.72×10^{-10}, and 1.67×10^{-9}, and the direction cosine equals one to displayed precision in every case. The effective correction rank is 17 rather than the nominal 30, consistent with the rank of the sample Jacobian at the initial state. Along the full trajectory, pointwise PB–SAV and damped Gauss–Newton agree in final objective to 1.24×10^{-11}. They also attain the same noise-free test MSE and final gradient norm in Table 2.

The pointwise components expose the individual residual Jacobians, so the PB–SAV correction reproduces the Gauss–Newton geometry up to the small shift regularization. Coarser partitions aggregate these directions before forming the pullback.

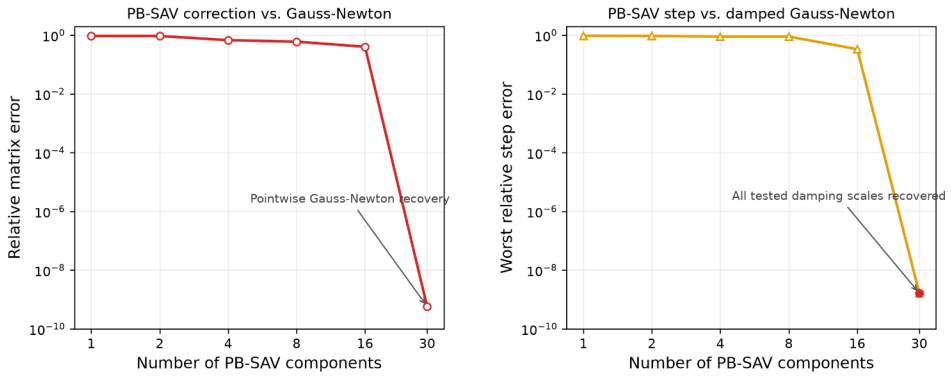

Figure 4: Fixed-state geometry for nonlinear regression. Left: relative error between the PB–SAV correction and the Gauss–Newton matrix. Right: the largest relative step error over $\eta = 0.1, 1, 10, 100$. Both errors collapse when each observation is represented by a separate component.

4.3. Operator learning: two-dimensional Darcy flow

We next apply PB–SAV with separate components for the supervised data loss and weight decay to operator learning and compare it with heavy-ball gradient descent and AdamW. For each permeability field $a(x, y)$, the target solution satisfies

$$-\nabla\cdot(a(x,y)\nabla u(x,y)) = 1, \qquad (x,y) \in \Omega = (0,1)^2, \qquad u|_{\partial\Omega} = 0. \tag{70}$$

The permeability is generated by thresholding a smoothed Gaussian random field with correlation length 0.12, which produces the two values 3 and 12. Finite-difference solutions are represented on a 33×33 grid. For each seed, the dataset contains 128 training functions and 64 independently generated test functions.

We use a 51,521-parameter convolutional-branch deep operator network (DeepONet) [3]. The branch encoder uses convolutional channel widths $8, 16, 32$, adaptive 4×4 pooling, and a 64-dimensional projection. The trunk is a three-hidden-layer tanh network of width 64 and maps spatial coordinates to the same latent dimension. Multiplication by $16x(1-x)y(1-y)$ enforces the homogeneous Dirichlet boundary condition exactly.

Let $N_{\mathrm{tr}} = 128$ and $N_x = 33^2$. For training field a_j, let $u_j(x_\ell)$ be the reference solution at grid point x_ℓ, and let $\widehat{u}_\theta(a_j; x_\ell)$ be the network prediction.

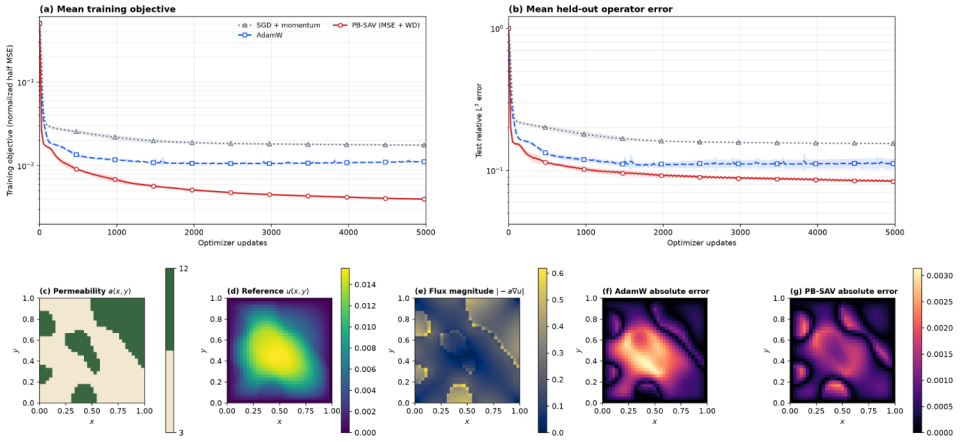

Figure 5: Five-seed Darcy DeepONet comparison. (a) Mean common training data loss and (b) mean held-out relative L^2 error; shaded regions show one sample standard deviation. The legend entry "SGD + momentum" in (a) and (b) is the full-batch heavy-ball gradient descent of Table 3. The lower row shows a fixed test sample selected by the maximum variation of the reference field: (c) permeability, (d) reference solution, (e) flux magnitude, (f) AdamW absolute error, and (g) PB–SAV absolute error.

Define

$$E_{\mathrm{data}}(\theta) = \frac{1}{2N_{\mathrm{tr}}N_x} \sum_{j=1}^{N_{\mathrm{tr}}} \sum_{\ell=1}^{N_x} \left(\widehat{u}_\theta(a_j; x_\ell) - u_j(x_\ell)\right)^2, \tag{71}$$

$$E_{\mathrm{wd}}(\theta) = \frac{\lambda}{2} \|\theta\|^2, \qquad \lambda = 3 \times 10^{-5}, \tag{72}$$

PB–SAV uses the two-component objective

$$F_{\mathrm{Darcy}}(\theta) = E_{\mathrm{data}}(\theta) + E_{\mathrm{wd}}(\theta). \tag{73}$$

The common training quantity reported here is E_{data}. Generalization is measured by the relative L^2 error on the held-out operator dataset.

The selected learning rates are 10^{-2}, 10^{-3}, and 3×10^{-3} for heavy-ball gradient descent, AdamW, and PB–SAV, respectively. PB–SAV uses $\beta = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, $\alpha = 0.5$, and $\rho^n \equiv 1$. These PB–SAV parameters are also used in the following Burgers experiments unless otherwise stated. Each method runs for 5,000 full-batch updates using the paired seeds $42, 101, 211, 307, 401$. The displayed field is the fixed test sample with the greatest variation in the reference solution.

PB–SAV attains the lowest training and test errors in Fig. 5 and Table 3. Relative to AdamW, it reduces the mean training data loss by 64.5% and

Table 3: Darcy DeepONet results after 5,000 updates over five paired seeds. Losses and errors are reported as mean ± sample standard deviation. Wall time is the mean recorded time for the full training run. Best mean training and test values are bold; wall time is reported for reference and is not ranked.

Method	Components	η	Train data loss	Test relative L^2	Wall time (s)
Heavy-ball gradient descent	–	10^{-2}	0.01771 ± 0.00041	0.15510 ± 0.00095	37.4
AdamW	–	10^{-3}	0.01125 ± 0.00024	0.11264 ± 0.01001	36.5
PB–SAV (data + weight decay)	2	3×10^{-3}	$\mathbf{0.00399 \pm 0.00011}$	$\mathbf{0.08512 \pm 0.00225}$	74.5

the mean held-out error by 24.4%. Relative to heavy-ball gradient descent, the reductions are 77.5% and 45.1%, respectively. The standard deviation of the PB–SAV test error is 0.00225, compared with 0.01001 for AdamW. In the lower row of Fig. 5, the PB–SAV prediction has a smaller error over most of the domain for the displayed test sample.

With two components, PB–SAV gives lower mean training and test errors than the two baselines at approximately twice the recorded wall time of AdamW. This comparison evaluates the complete optimizer. The following forward Burgers experiment includes a one-component PB–SAV baseline to examine the effect of the decomposition itself.

4.4. Forward Burgers PINN: one versus four loss components

The forward problem compares one aggregated component with four resolved components at $\alpha = 1/2$, under the same momentum and adaptive scaling rules. The inverse problem then adds sparse observations and an unknown viscosity.

We consider the viscous Burgers equation

$$u_t + uu_x - \nu u_{xx} = 0, \qquad (x,t) \in [-1,1] \times [0,1], \qquad \nu = 0.01, \tag{74}$$

with initial and boundary conditions

$$u(x,0) = -\sin(\pi x), \qquad u(-1,t) = u(1,t) = 0. \tag{75}$$

The PINN is a tanh network with six hidden layers of width 64 and 21,057 trainable parameters. Every run uses 10,000 fixed scrambled Sobol collocation points [30, 31], 256 initial points, 256 time points on each boundary, and 10,000 independent validation collocation points. The relative solution error is evaluated against a grid-refined backward differentiation formula (BDF) method-of-lines reference on a 256×101 space–time grid.

Let $r_f = u_t + uu_x - \nu u_{xx}$. For sampled values $w_1, \dots, w_N$, we use $\mathrm{MSE}(w) = N^{-1} \sum_{j=1}^{N} w_j^2$. The common PINN task objective is

$$E_{\mathrm{res}} = \frac{1}{2} \mathrm{MSE}(r_f), \tag{76a}$$

$$E_{\mathrm{bc}} = \frac{1}{4} \left[\mathrm{MSE}(u(-1,t)) + \mathrm{MSE}(u(1,t)) \right], \tag{76b}$$

$$E_{\mathrm{ic}} = \frac{1}{2} \mathrm{MSE}(u(x,0) + \sin(\pi x)), \tag{76c}$$

$$F_{\mathrm{task}} = E_{\mathrm{res}} + E_{\mathrm{bc}} + E_{\mathrm{ic}}. \tag{76d}$$

All methods use the weight-decay coefficient $\lambda = 10^{-6}$. For PB–SAV, weight decay is represented by the explicit component

$$F_{\mathrm{reg}}(\theta) = F_{\mathrm{task}}(\theta) + E_{\mathrm{wd}}(\theta), \qquad E_{\mathrm{wd}}(\theta) = \frac{\lambda}{2} \|\theta\|^2 . \tag{77}$$

We report F_{task}, which is common to all methods, as the primary optimization quantity.

We compare heavy-ball gradient descent, AdamW, adaptive PB–SAV with momentum using one aggregate component, and the same PB–SAV method using the four components E_{res}, E_{bc}, E_{ic}, and E_{wd}. The total SAV shift is 10^{-12}. One quarter is assigned to weight decay, and the remainder is distributed among the three physical terms in proportion to their sample counts. A seed-0 pilot selects learning rates from the task objective only. The selected rates are 10^{-2} for heavy-ball gradient descent and 10^{-3} for AdamW and both PB–SAV variants. Reported runs use 10,000 updates at the paired seeds $42, 2273, 2669, 5439, 8041$.

The mean wall times were 316.6 ± 8.4, 328.8 ± 6.1, 491.8 ± 6.5, and 496.9 ± 16.0 seconds for heavy-ball gradient descent, AdamW, one-component PB–SAV, and four-component PB–SAV, respectively.

The four-component PB–SAV method attains the lowest task objective, the lowest solution error, and the smallest late-stage oscillation in Fig. 6 and Table 4. Relative to one-component PB–SAV, it reduces the mean tail objective by 64.7%, the mean final solution error by 50.2%, and the mean tail CV from 58.9% to 5.4%. Relative to AdamW, the reductions in task objective and solution error are 80.7% and 82.0%, respectively. Relative to heavy-ball gradient descent, they are 87.9% and 76.1%.

Splitting the objective into four components costs 1.0% of the recorded wall time over the one-component run, once the common PINN forward graph has been evaluated; the one- and four-component PB–SAV runs are respectively 49.6% and 51.1% slower than AdamW.

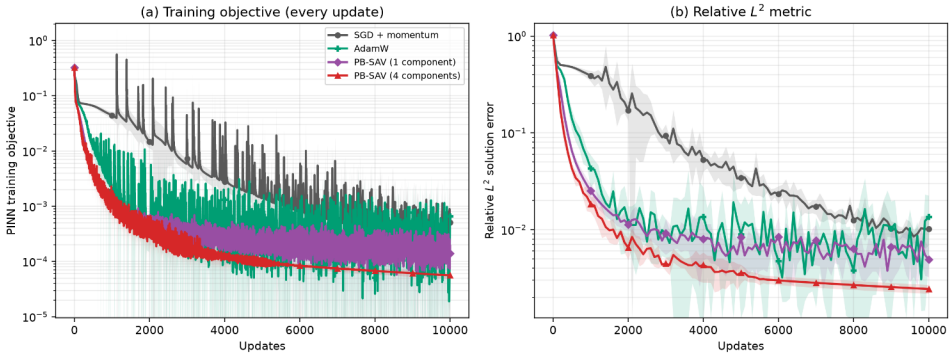

Figure 6: Five-seed comparison for the forward Burgers PINN. (a) Mean common task objective, recorded at every update, and (b) mean relative L^2 solution error. Shaded regions show one sample standard deviation.

Both runs form the same component gradients. The one-component run supplies a single aggregated component, so its curvature gap vanishes and its correction is the aggregate B_1^n for every α; the four-component run has $m = 4$ and $\alpha = 1/2$, so Eq. (13) gives $B_\alpha^n = \frac{1}{2}(B_1^n + B_m^n)$. The two runs use the same architecture, samples, mobility update rule, momentum parameters, and learning rate. Keeping the residual, boundary, initial, and regularization directions separate produces a smoother late optimization trajectory and a more accurate PDE solution in these runs.

4.5. Inverse Burgers PINN: viscosity identification

The final experiment uses the same Burgers equation and prescribed conditions, but treats the viscosity as unknown. The reference value is $\nu_* = 0.01$, and optimization starts from $\nu^0 = 0.02$. We optimize $\xi = \log \nu$, which enforces $\nu > 0$. Let $\theta_{\text{net}} \in \mathbb{R}^{21057}$ denote the weights of the PINN with six hidden layers of width 64. The full optimization state is $z = (\theta_{\text{net}}, \xi) \in \mathbb{R}^{21058}$, with $\nu = e^\xi$.

Every run uses 10,000 fixed collocation points, 256 initial points, 256 points on each boundary, and 100 noiseless interior observations sampled from the validated reference solution. An independent set of 10,000 collocation points is used for validation. The relative solution error is evaluated on the same 256×101 grid used in the forward problem.

The residual, boundary, and initial energies are defined as in Eq. (76), with the trainable viscosity inserted into the residual. Let $(x_j^{\text{obs}}, t_j^{\text{obs}})$ be the observation sites and y_j the corresponding reference values. The observation

Table 4: Forward Burgers results after 10,000 updates over five paired seeds. For each seed, the tail mean and coefficient of variation (CV) are computed over the final 2,000 updates; the table reports their mean ± sample standard deviation across seeds. Here $\mathrm{CV} = 100 s_{\mathrm{tail}} / \bar{F}_{\mathrm{tail}}$. Best mean values are bold.

Method	η	Tail task objective	Tail objective CV
Heavy-ball GD	0.01	$(5.095 \pm 0.660) \times 10^{-4}$	$130.2\% \pm 22.0\%$
AdamW	0.001	$(3.181 \pm 0.450) \times 10^{-4}$	$161.5\% \pm 22.0\%$
PB–SAV (1 component)	0.001	$(1.741 \pm 0.430) \times 10^{-4}$	$58.9\% \pm 31.0\%$
PB–SAV (4 components)	0.001	$\mathbf{(6.154 \pm 0.840) \times 10^{-5}}$	$\mathbf{5.4\% \pm 0.3\%}$

Method	Final relative L^2 error
Heavy-ball GD	$(1.018 \pm 0.330) \times 10^{-2}$
AdamW	$(1.356 \pm 0.890) \times 10^{-2}$
PB–SAV (1 component)	$(4.893 \pm 1.700) \times 10^{-3}$
PB–SAV (4 components)	$\mathbf{(2.435 \pm 0.260) \times 10^{-3}}$

loss is

$$E_{\mathrm{data}} = \frac{1}{200} \sum_{j=1}^{100} \left(u_{\theta_{\mathrm{net}}}(x_j^{\mathrm{obs}}, t_j^{\mathrm{obs}}) - y_j \right)^2 . \tag{78}$$

The common inverse-task objective and the PB–SAV regularized objective are

$$F_{\mathrm{inv}} = E_{\mathrm{res}} + E_{\mathrm{bc}} + E_{\mathrm{ic}} + E_{\mathrm{data}}, \qquad F_{\mathrm{inv}}^{\lambda} = F_{\mathrm{inv}} + \frac{\lambda}{2} \left\| \theta_{\mathrm{net}} \right\|^2 . \tag{79}$$

The regularizer acts only on the network parameters. The viscosity parameter is identified through the physics and observation terms.

The one-component PB–SAV method aggregates all task and regularization terms. The four-component formulation uses E_{res}, $E_{\mathrm{bc}} + E_{\mathrm{ic}}$, E_{data}, and E_{wd}. The boundary and initial terms are grouped as prescribed-condition losses, while the residual, observation, and regularization directions remain separate. Component shifts are additive under aggregation. Of the total shift 10^{-12}, 25% is assigned to regularization.

The pilot described below selects the weight-decay coefficient $\lambda = 3 \times 10^{-7}$. We report F_{inv}, which is common to all methods.

A separate 500-update pilot on seed 0 selects learning rates using the tail training objective with a penalty for late objective increases. The selected rates are 10^{-2}, 10^{-3}, 3×10^{-3}, and 10^{-2} for heavy-ball gradient descent, AdamW, one-component PB–SAV, and four-component PB–SAV, respectively. Reported runs use 10,000 updates at the same five seeds as the

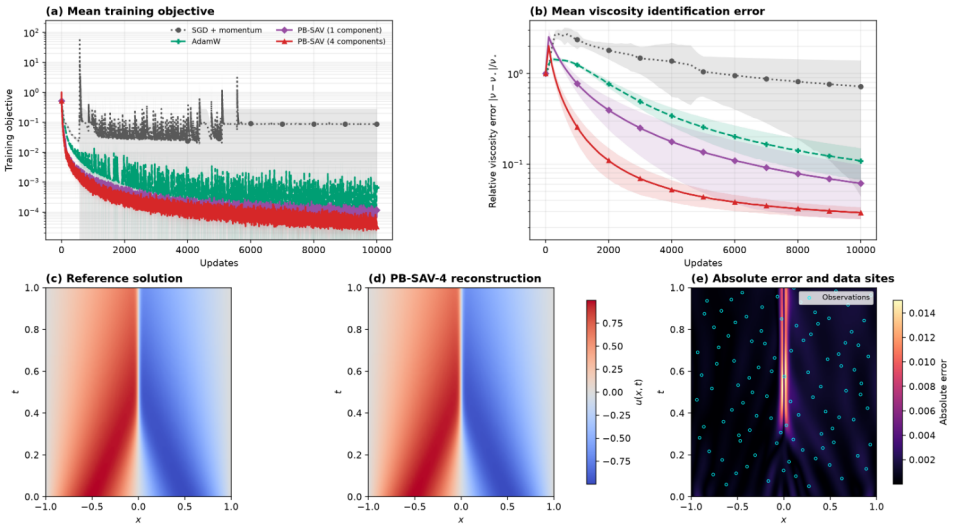

Figure 7: Five-seed comparison for inverse Burgers identification. (a) Mean common task objective and (b) mean relative viscosity error; shaded regions show one sample standard deviation. The lower row uses seed 42, the median four-component PB–SAV seed according to final viscosity error: (c) reference solution, (d) four-component PB–SAV reconstruction, and (e) absolute error with the 100 observation locations.

forward experiment.

The mean wall times were 404.8 ± 9.2, 408.2 ± 8.2, 567.1 ± 33.0, and 583.6 ± 35.0 seconds for heavy-ball gradient descent, AdamW, one-component PB–SAV, and four-component PB–SAV, respectively. All four methods use $\lambda = 3 \times 10^{-7}$.

Four-component PB–SAV gives the best mean result for the training objective, viscosity identification, and solution reconstruction in Fig. 7 and Table 5. Relative to one-component PB–SAV at its own selected learning rate, it reduces the mean tail objective by 49.3%, the mean viscosity error by 52.5%, and the mean solution error by 55.1%. Its solution-error standard deviation is approximately 10.6 times smaller, and its viscosity-error standard deviation is approximately 8.6 times smaller. Relative to AdamW, the mean reductions in these three quantities are 86.3%, 73.2%, and 76.7%, respectively. The four-component method is 2.9% slower than one-component PB–SAV in the recorded wall time and 43.0% slower than AdamW.

Heavy-ball gradient descent converges to a near-zero field on seed 2669, which accounts for the large mean and variance in its aggregate row. AdamW approaches the viscosity on all five seeds, but its task objective exhibits large

Table 5: Inverse Burgers results after 10,000 updates over five paired seeds. Values are mean ± sample standard deviation. For each seed, the tail task objective is averaged over the final 2,000 updates of F_{inv}. Best mean values are bold.

Method	η	Tail task objective
Heavy-ball GD	0.01	$(0.878 \pm 1.900) \times 10^{-1}$
AdamW	0.001	$(3.515 \pm 0.330) \times 10^{-4}$
PB–SAV (1 component)	0.003	$(9.507 \pm 2.000) \times 10^{-5}$
PB–SAV (4 components)	0.01	$\mathbf{(4.816 \pm 2.000) \times 10^{-5}}$

Method	Final viscosity relative error	Final relative L^2 error
Heavy-ball GD	$(7.213 \pm 6.800) \times 10^{-1}$	$(2.293 \pm 4.300) \times 10^{-1}$
AdamW	$(1.088 \pm 0.420) \times 10^{-1}$	$(1.440 \pm 0.650) \times 10^{-2}$
PB–SAV (1 component)	$(6.141 \pm 3.700) \times 10^{-2}$	$(7.476 \pm 3.300) \times 10^{-3}$
PB–SAV (4 components)	$\mathbf{(2.914 \pm 0.430) \times 10^{-2}}$	$\mathbf{(3.356 \pm 0.310) \times 10^{-3}}$

late spikes.

The representative reconstruction in Fig. 7 tracks the reference solution throughout most of the space–time domain. The largest remaining error occurs near the steep central layer, where the solution is most sensitive to viscosity. The four-component optimizer attains lower mean errors at its selected learning rate with a small additional cost over one-component PB–SAV; the forward experiment compares the pullbacks at a common learning rate.

5. Conclusions

We placed the PB–SAV component correction inside an optimizer with momentum and an adaptive mobility, applying it to the gradient and the stored momentum in a single implicit solve. The update dissipates an exact modified energy for every mobility that is nonincreasing in the Loewner order, and its local stability at a stationary point is governed by the Hessian minus twice the correction. The aggregate correction vanishes at such a point while the component correction need not, so resolving the objective into components can extend the admissible step sizes. Each iteration adds one dense solve of order at most the number of components.

The quadratic and regression tests show that the correction can reproduce a reference increment while its error as a matrix remains large. In the forward Burgers comparison, four components reduce the mean tail objective by 64.7% and the final solution error by 50.2% relative to one component at the same

learning rate and momentum settings. How to choose the components when the objective is not already split, and how to do so without one gradient evaluation per component, remain open.

Declaration of competing interest

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data and code availability

Code and data will be made available in a public repository on publication.

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Appendix A. Convergence proofs

A.1. Proof of the modified energy law

Proof of Theorem 3.6. Take the inner product of Eq. (28) with $\Delta\theta^{n+1} = M^{n+1}p^{n+1}$. Weighted polarization gives

$$\begin{aligned}&\frac{(p^{n+1})^{\mathsf{T}} M^{n+1}(p^{n+1} - \beta p^n)}{\eta} \\ &\quad = \frac{1}{2\eta}\left(\left\|p^{n+1}\right\|_{M^{n+1}}^2 - \beta^2 \left\|p^n\right\|_{M^{n+1}}^2 + \left\|p^{n+1} - \beta p^n\right\|_{M^{n+1}}^2\right). \qquad \text{(A.1)}\end{aligned}$$

Insert and subtract $\left\|p^n\right\|_{M^n}^2/(2\eta)$ to obtain

$$\begin{aligned}&\frac{\left\|p^{n+1}\right\|_{M^{n+1}}^2 - \beta^2 \left\|p^n\right\|_{M^{n+1}}^2}{2\eta} \\ &\quad = \frac{\left\|p^{n+1}\right\|_{M^{n+1}}^2 - \left\|p^n\right\|_{M^n}^2}{2\eta} + \frac{\left\|p^n\right\|_{M^n - M^{n+1}}^2}{2\eta} + \frac{1-\beta^2}{2\eta}\left\|p^n\right\|_{M^{n+1}}^2.\end{aligned}$$

Substitute the scalar and pullback identities in Eq. (18). Collecting the scalar and kinetic energy changes yields Eq. (39). Each remaining term is nonnegative because $S_m^n \succeq 0$, $M^n - M^{n+1} \succeq 0$, and $0 \le \beta < 1$. □

A.2. Proof of the local stability criterion

Proof of Theorem 3.9. Write $\widehat{H}_* = M^{1/2} H_* M^{1/2}$ and $\widehat{B}_* = M^{1/2} B_* M^{1/2}$. For the linearized variables $x^n = M^{-1/2}(\theta^n - \theta_*)$ and $w^n = M^{1/2} p^n$,

$$(I_d + \eta \widehat{B}_*) w^{n+1} = \beta w^n - \eta \kappa_* \widehat{H}_* x^n, \qquad x^{n+1} = x^n + w^{n+1}. \tag{A.2}$$

Derivatives of the inverse coefficient multiply a zero right-hand side, and variations of q/Q multiply the zero gradient. The invertible change of state from (x^n, w^n) to $(x^n, x^n - w^n)$ gives the matrix polynomial

$$\begin{aligned} \mathcal{P}(z) = z^2 (I_d + \eta \widehat{B}_*) \\ - z\big((1+\beta) I_d + \eta \widehat{B}_* - \eta \kappa_* \widehat{H}_*\big) + \beta I_d. \end{aligned} \tag{A.3}$$

Its determinant vanishes exactly at the eigenvalues of the state map, including when $\beta = 0$.

If $\mathcal{P}(z)\xi = 0$ with $\xi \in \mathbb{C}^d \setminus \{0\}$, multiplication by ξ^* gives the scalar quadratic

$$(1 + \eta b) z^2 - (1 + \beta + \eta b - \eta \kappa_* h) z + \beta = 0, \tag{A.4}$$

where $b = \xi^* \widehat{B}_* \xi / (\xi^* \xi) \ge 0$ and $h = \xi^* \widehat{H}_* \xi / (\xi^* \xi) > 0$ are real. The leading coefficient exceeds β. The values of this polynomial at 1 and -1 are $\eta \kappa_* h$ and $2(1+\beta) + 2\eta b - \eta \kappa_* h$, respectively, both positive under Eq. (52). Nonreal roots have squared modulus $\beta/(1+\eta b) < 1$. For real roots, the same root product and the positive values at 1 and -1 exclude roots outside $(-1, 1)$. This proves sufficiency.

Conversely,

$$\mathcal{P}(-1) = 2(1+\beta) I_d + 2\eta \widehat{B}_* - \eta \kappa_* \widehat{H}_*.$$

If this matrix is singular, -1 is an eigenvalue. If it has a negative eigenvalue, $\mathcal{P}(-t)$ becomes positive definite for sufficiently large t, since its leading coefficient is positive definite. Continuity of its smallest eigenvalue gives a singular matrix $\mathcal{P}(-t)$ at some real $t > 1$, and hence a state eigenvalue $-t < -1$. This proves necessity. □

A.3. Comparison with the appended recurrence

On a common invariant one-dimensional mode of $\widehat{H}_*$ and $\widehat{B}_*$, let $h > 0$ and $b \ge 0$ be their curvatures. If $b = \kappa_* h$, the characteristic equation reduces to

$$(1 + \eta b) z^2 - (1 + \beta) z + \beta = 0.$$

For $\beta > 0$ and $\eta b > (1-\beta)^2/(4\beta)$, its roots have modulus $\sqrt{\beta/(1+\eta b)}$, which tends to zero as $\eta b \to \infty$. For $\beta = 0$, the roots are 0 and $1/(1+\eta b)$. The frozen appended recurrence has root product β, so its spectral radius is at least $\sqrt{\beta}$. On modes with large ηb the momentum update therefore has the smaller spectral radius; this is a stiff-limit statement, not a uniform rate comparison. Its stability condition, however, is

$$\eta \kappa_* H_* \prec 2(1+\beta)(M^{-1} + \eta B_*),$$

which admits at least the interval of the momentum update for the same pullback. It follows by diagonalizing the positive definite generalized Hessian pair $(H_*, M^{-1} + \eta B_*)$. The interval enlargement in Corollary 3.10 compares component and aggregate pullbacks within the momentum update.

A.4. Local convergence with scalar relaxation

Proof of Theorem 3.11. Set $z = (\theta - \theta_*, p)$ and $Q_* = Q(\theta_*)$. Let $T(q, z)$ be the parameter–momentum update, without applying scalar relaxation. This map is continuously differentiable near $(Q_*, 0)$, and $T(q, 0) = 0$ for every nearby q. Its derivative $L = D_z T(Q_*, 0)$ is Schur stable by Theorem 3.9. The convergent series

$$P = \sum_{j=0}^{\infty} (L^{\mathsf{T}})^j L^j$$

defines a positive definite matrix with $L^{\mathsf{T}} P L = P - I$. Hence $\|L\|_P < 1$. Continuity of $D_z T$ and integration along the segment from 0 to z give a cylinder

$$\|z\|_P \le a_z, \qquad |q - Q_*| \le a_q < Q_*/2$$

and $r \in (0,1)$ such that

$$\|T(q,z)\|_P \le r \, \|z\|_P \,. \tag{A.5}$$

The scalar must remain in this cylinder. Initialization and the acceptance rule give the invariant $0 \le q^n \le Q(\theta^n)$. Within the cylinder, uniformly in $\rho \in [0,1]$, the increment of Eq. (27) and the scalar increment d of Eq. (17) satisfy

$$\|\Delta\| + \left\|p^+\right\| = O(\|z\|_P), \qquad d = O(\|z\|_P^2), \qquad 0 \le \mathcal{D} = O(\|z\|_P^2).$$

Write the two branches of Eq. (45) as $q_{\mathrm{bud}}^2 = (q+d)^2 + \rho \mathcal{D}$ and $q_{\mathrm{cap}}^2 = Q(\theta + \Delta)^2$. Then

$$|q_{\mathrm{bud}}^2 - q^2| \le c_4 \, \|z\|_P^2 \,.$$

Since $q \le Q(\theta)$ and $\nabla F(\theta_*) = 0$,

$$q_{\mathrm{cap}}^2 - q^2 \ge F(\theta + \Delta) - F(\theta) \ge -c_5 \, \|z\|_P^2 \,.$$

Both branches are bounded below by $q^2 - c_6 \|z\|_P^2$ with $c_6 = \max\{c_4, c_5\}$, while the budget branch bounds their minimum above. Thus

$$|(q^+)^2 - q^2| \le c_6 \|z\|_P^2 .$$

Dividing by $q^+ + q \ge q \ge Q_*/2$ gives

$$|q^+ - q| \le c_2 \|z\|_P^2 . \tag{A.6}$$

This estimate covers either selected branch and branch ties.

Because $\nabla Q(\theta_*) = 0$, $|q^0 - Q_*| \le c_3 \left\|z^0\right\|_P^2$. Choose the initial neighborhood so that $\left\|z^0\right\|_P < a_z$ and

$$\left(c_3 + \frac{c_2}{1 - r^2}\right) \left\|z^0\right\|_P^2 < a_q.$$

Induction using Eqs. (A.5) and (A.6) then gives

$$\|z^n\|_P \le r^n \left\|z^0\right\|_P , \qquad |q^n - Q_*| \le \left(c_3 + \frac{c_2}{1 - r^2}\right) \left\|z^0\right\|_P^2 < a_q.$$

The iterates cannot leave the cylinder, closing the induction. Scalar increments are summable, so $q^n \to q_\infty \ge Q_*/2$. The invariant $q^n \le Q(\theta^n)$ gives $q_\infty \le Q_*$, and the drift bound gives Eq. (58). Equivalence of norms yields Eq. (57). □

A.5. Global stationarity estimates

With $L_B = \sup_n \|B_\alpha^n\|$, the constants of Corollary 3.14 are

$$\begin{aligned} C_{\mathrm{mom}} &= \frac{4\eta}{\underline{\kappa}^2 \mu_{\min}} \left[\left(\eta^{-1} + \mu_{\max} L_B\right)^2 + \frac{\beta^2 \mu_{\max}^2 L_B^2}{1 - \beta^2} \right], \\ C_{\mathrm{dir}} &= \frac{\eta \left(\eta^{-1} + L_B\right)^2}{\underline{\kappa}^2}. \end{aligned} \tag{A.7}$$

One sets $K = C_{\mathrm{mom}}$ under the assumptions of Theorem 3.12 and $K = C_{\mathrm{dir}}$ under those of Corollary 3.13.

Proof of Theorem 3.12. Sum Eq. (46) and use Eq. (59) and $\mathcal{H}^n \ge 0$ to obtain $\sum_n \mathcal{D}^n < \infty$. The damping and inertial residual terms in Eq. (39), together with the uniform lower mobility bound, imply $p^n \to 0$ and $p^{n+1} - \beta p^n \to 0$. The uniform upper bound gives $\Delta\theta^{n+1} = M^{n+1} p^{n+1} \to 0$. Using these limits and the boundedness of B_α^n in Eq. (28) yields $(q^n/Q^n) g^n \to 0$, and the ratio lower bound gives $g^n \to 0$. Since $F \in C^1$, every accumulation point is stationary. □

Proof of Corollary 3.13. The direct update law Eq. (20) and summability give $\Delta\theta^{n+1} \to 0$. The state equation Eq. (19a), the bounded pullback, and the ratio lower bound then imply $g^n \to 0$. The accumulation-point claim follows by continuity of ∇F. □

Proof of Corollary 3.14. For the momentum update, set $e^n = p^{n+1} - \beta p^n$. The state equation, $p^{n+1} = e^n + \beta p^n$, and the residual and damping terms in Eq. (39) give

$$\begin{aligned} \frac{q^n}{Q^n} g^n &= -\frac{e^n}{\eta} - B_\alpha^n M^{n+1}(e^n + \beta p^n), \\ \|g^n\| &\le \frac{1}{\underline{\kappa}} \left[\left(\eta^{-1} + \mu_{\max} L_B\right) \|e^n\| + \beta \mu_{\max} L_B \|p^n\| \right], \\ \mathcal{D}^n &\ge \frac{\mu_{\min}}{2\eta} \left(\|e^n\|^2 + (1-\beta^2) \|p^n\|^2 \right). \end{aligned} \tag{A.8}$$

The inequality $(a+b)^2 \le 2a^2 + 2b^2$ gives $\|g^n\|^2 \le C_{\mathrm{mom}} \mathcal{D}^n$.

For the direct update, Eq. (19a) gives

$$\|g^n\| \le \frac{\eta^{-1} + L_B}{\underline{\kappa}} \left\|\Delta\theta^{n+1}\right\|, \qquad \mathcal{D}^n \ge \frac{\left\|\Delta\theta^{n+1}\right\|^2}{\eta}, \tag{A.9}$$

and therefore $\|g^n\|^2 \le C_{\mathrm{dir}} \mathcal{D}^n$. Finally, summing Eq. (46) and using Eq. (59) gives $(1-\bar{\rho}) \sum_{n=0}^{N-1} \mathcal{D}^n \le \mathcal{H}^0$. This proves both estimates in Eq. (65); the infinite-series claim follows by letting $N \to \infty$. □

Appendix B. A curvature condition for exact scalar tracking

Full relaxation preserves exact scalar tracking when the correction covers the true objective remainder along each step, up to a fixed fraction of the inertial dissipation. Define

$$R_F^n = F(\theta^{n+1}) - F(\theta^n) - (g^n)^{\mathsf{T}} \Delta\theta^{n+1} \tag{B.1}$$

and the inertial part of the dissipation,

$$\begin{aligned} \mathcal{D}_{\mathrm{in}}^n = \frac{\|p^n\|_{M^n - M^{n+1}}^2}{2\eta} + \frac{\left\|p^{n+1} - \beta p^n\right\|_{M^{n+1}}^2}{2\eta} \\ + \frac{1-\beta^2}{2\eta} \|p^n\|_{M^{n+1}}^2 . \end{aligned} \tag{B.2}$$

The true objective with kinetic energy is

$$\mathcal{E}^n = F(\theta^n) + C + \frac{\|p^n\|_{M^n}^2}{2\eta}. \tag{B.3}$$

Proposition B.1. *At a consistent state $q^n = Q^n$, the momentum update satisfies*

$$\mathcal{E}^n - \mathcal{E}^{n+1} = \mathcal{D}_{\text{in}}^n + \left\|\Delta\theta^{n+1}\right\|_{B_\alpha^n}^2 - R_F^n. \tag{B.4}$$

Proof. Take the inner product of Eq. (28) with $\Delta\theta^{n+1} = M^{n+1}p^{n+1}$ and use weighted polarization. With $q^n/Q^n = 1$, this gives

$$\frac{\|p^n\|_{M^n}^2 - \left\|p^{n+1}\right\|_{M^{n+1}}^2}{2\eta} = \mathcal{D}_{\text{in}}^n + (g^n)^\mathsf{T}\Delta\theta^{n+1} + \left\|\Delta\theta^{n+1}\right\|_{B_\alpha^n}^2.$$

Subtract the true objective increment to obtain Eq. (B.4). □

Theorem B.2. *Fix $\eta > 0$, $0 \le \beta < 1$, and $0 \le \alpha \le 1$, and let $0 \prec M^{n+1} \preceq M^n$. Initialize $q^0 = Q(\theta^0)$ and use full relaxation $\rho^n \equiv 1$. Suppose that every step is well defined and satisfies*

$$R_F^n \le \left\|\Delta\theta^{n+1}\right\|_{B_\alpha^n}^2 + \gamma^n\mathcal{D}_{\text{in}}^n, \qquad 0 \le \gamma^n \le \bar{\gamma} < 1. \tag{B.5}$$

Then $q^n = Q(\theta^n)$ for every n, and

$$\mathcal{E}^n - \mathcal{E}^{n+1} \ge (1-\bar{\gamma})\mathcal{D}_{\text{in}}^n \ge 0. \tag{B.6}$$

If also $\mu_{\min} I_d \preceq M^n \preceq \mu_{\max} I_d$ for constants $0 < \mu_{\min} \le \mu_{\max} < \infty$, and $\sup_n \|B_\alpha^n\| < \infty$, then $p^n \to 0$, $\Delta\theta^{n+1} \to 0$, and $\nabla F(\theta^n) \to 0$.

Proof. Suppose $q^n = Q^n$. Combining the modified and true energy identities gives

$$(\bar{q}^{n+1})^2 + \mathcal{D}^n - Q(\theta^{n+1})^2 = \mathcal{D}_{\text{in}}^n + \left\|\Delta\theta^{n+1}\right\|_{B_\alpha^n}^2 - R_F^n \ge 0. \tag{B.7}$$

Thus the minimum in Eq. (45) selects $Q(\theta^{n+1})^2$. Induction proves exact tracking, and Eq. (B.4) yields Eq. (B.6).

Since $\mathcal{E}^n \ge 0$, summing the energy law gives $\sum_n \mathcal{D}_{\text{in}}^n < \infty$. The damping and residual terms, with the uniform lower mobility bound and $\beta < 1$, imply $p^n \to 0$ and $p^{n+1} - \beta p^n \to 0$. The upper bound gives $\Delta\theta^{n+1} = M^{n+1}p^{n+1} \to 0$. Finally, Eq. (28), exact tracking, and the bounded pullback imply $g^n \to 0$. □

For a quadratic objective, the condition with $\gamma^n = 0$ becomes

$$\frac{1}{2}(\Delta\theta^{n+1})^\mathsf{T} H \Delta\theta^{n+1} \le \left\|\Delta\theta^{n+1}\right\|_{B_\alpha^n}^2.$$

The correction must cover half the Hessian quadratic form along the step, not dominate the Hessian in every direction. For nonlinear least squares, write $\nabla^2 F(\theta^n) = G^n + A^n$, where G^n is the residual Gauss–Newton matrix. If

$\|A^n\| \le \varepsilon^n$ and the Hessian is Lipschitz along the step segment with constant L^n, Taylor's formula gives

$$R_F^n \le \frac{1}{2}\left\|\Delta\theta^{n+1}\right\|_{G^n}^2 + \frac{\varepsilon^n}{2}\left\|\Delta\theta^{n+1}\right\|^2 + \frac{L^n}{6}\left\|\Delta\theta^{n+1}\right\|^3.$$

Consequently, a sufficient condition is

$$(\Delta\theta^{n+1})^{\mathsf{T}}\left(B_\alpha^n - \frac{1}{2}G^n\right)\Delta\theta^{n+1} \ge \frac{\varepsilon^n}{2}\left\|\Delta\theta^{n+1}\right\|^2 + \frac{L^n}{6}\left\|\Delta\theta^{n+1}\right\|^3. \quad \text{(B.8)}$$

The projection identity Eq. (16) describes the curvature omitted by the least-squares components. Neither that identity nor modified energy dissipation alone imposes Eq. (B.5), which can instead be checked from the objective increment without forming a Hessian.

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